

MA 374 (2026): Financial Engineering Lab

Lab 01: Binomial Pricing of European Options

1 Objective

The objective of this experiment is to price a European call option and a European put option using the binomial option pricing model under the risk-neutral framework. The convergence of the binomial prices is analyzed by varying the number of time steps, and the evolution of option prices over time is also studied.

2 Model Parameters

The parameters used in this experiment are:

- Initial stock price: $S(0) = 9$
- Strike price: $K = 10$
- Time to maturity: $T = 3$ years
- Risk-free interest rate (continuous compounding): $r = 0.06$
- Volatility: $\sigma = 0.3$

The time step is defined as $\Delta t = \frac{T}{M}$, where M is the number of time steps. The up and down factors are given by:

$$u = e^{\sigma\sqrt{\Delta t} + (r - \frac{1}{2}\sigma^2)\Delta t}, \quad d = e^{-\sigma\sqrt{\Delta t} + (r - \frac{1}{2}\sigma^2)\Delta t}$$

The risk-neutral probability is:

$$\tilde{p} = \frac{e^{r\Delta t} - d}{u - d}$$

The model proceeds only when the no-arbitrage condition $0 < \tilde{p} < 1$ is satisfied.

3 Part (A): Option Prices for Different Values of M

The prices of the European call and put options at time $t = 0$ were computed for various values of M .

Table 1: Option Prices at $t = 0$ for Different Values of M

M	Call Price	Put Price
1	2.463367	1.816069
5	2.183245	1.535947
10	2.124835	1.477538
20	2.129583	1.482285
50	2.127215	1.479917
100	2.123667	1.476369
200	2.120209	1.472912

4 Part (B): Convergence of Option Prices

The option prices were computed for values of M ranging from 1 to 200, using step sizes of 1 and 5. The corresponding plots of option prices versus M were generated.

Observations

- For small values of M , both call and put option prices exhibit noticeable oscillations.
- As M increases, these oscillations reduce and the prices stabilize.
- The call option price converges to approximately 2.12, while the put option price converges to approximately 1.47.
- This behavior confirms the convergence of the binomial pricing model to its continuous-time limit as $M \rightarrow \infty$.

5 Part (C): Option Prices at Different Times for $M = 20$

For $M = 20$, the option prices were evaluated at different times before maturity.

Table 2: Option Prices at Different Times for $M = 20$

Time t	Call Price	Put Price
0.00	2.129583	1.482285
0.30	2.073440	1.414385
0.75	1.997121	1.320029
1.50	1.739074	1.030817
2.70	1.147463	0.386331

Observations

- Both call and put option prices decrease as time increases due to the reduction in time to maturity.
- The put option price decreases more rapidly as the probability of finishing in-the-money declines.
- Near maturity, the option prices approach their intrinsic values.

6 Conclusion

In this experiment, the binomial option pricing model was implemented to price European call and put options. The results demonstrate the convergence of option prices as the number of time steps increases and illustrate the time decay of option values. The binomial model provides an effective discrete-time approximation to continuous-time option pricing models.